



Louis Ghiglione

Seeking a 12-week International Mobility – Engineering Degree ENSEEIHT (3EA)

About Me

21-year-old engineering student at ENSEEIHT and apprentice microelectronics engineer. Dynamic with strong international volunteering experience, I am looking for a 12-week international mobility. Driving license (B).

Contact

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Skills

- **Microelectronics & Electronics:** Design of complex analog and digital circuits, CMOS technology with CADENCE, VHDL (Quartus, Vivado), FPGA (Xilinx), ASIC (synthesis, placement and routing), PCB retrofit and routing (Eagle, EasyEDA, KiCad), Microcap.
- **Electrical & Power Electronics:** Wiring, converters, B1V and B2V electrical safety clearances.
- **Automation & IT:** Python, C/C++, Arduino, 1st and 2nd order systems, MATLAB/Simulink, TIA portal, Control Expert, Grafcet/LD/ST, GEMMA, LabVIEW, Codesys, WAGO/eCockpit.
- **Web & IT:** HTML/CSS/JS, MySQL, Visual Basic.
- **Networking:** Applied basics (Cisco CCNA).

Languages

- **French:** Native
- **English:** Intermediate (B1 - TOEIC 615)
- **Spanish:** Basic (A2)

Interests

- Maker Culture (Home automation, DIY, technological reappropriation).
- Climbing (6B+), Hiking, and Mountain Biking.
- Reading.

Professional Experience

2026 - Present: Apprentice Analog/Digital Microelectronics Design Engineer - ISAE Supaero (Toulouse)

- Integrated into the Integrated Image Sensors (CIMI) research group.
- Design of readout circuit architectures in CMOS microelectronics technology using CADENCE.
- Hardware architecture definition and implementation of control routines in VHDL, with verification on FPGA (Xilinx, Vivado).
- ASIC design including synthesis, placement and routing, as well as verification.

2025 - 2026: Automation Engineering Work-Study & Intern - Fare ingenierie

- Design and integration of PLC programs and implementation of supervision systems.
- R&D development of an end-to-end IoT architecture: integration of LoRaWAN sensors via MQTT protocol on PLCs, specifically safety sensors for the ITER nuclear site.
- Development of a secure web backend (Python, FastAPI, Docker) and MySQL database management.

Technical Projects

- **Park4Move Project (LoRaWAN):** Development of a website and connected urban sensors. European "Service Learning" award winner.
- **Robotics:** Participation in 3 national competitions (design, prototyping, and maintenance).

Volunteering & Leadership

- **Scouting (10 years of experience):** Unit Leader (design of educational projects, team management). Fully autonomous participation in the Roverway 2024 European scout gathering in Norway.
- **Student Life:** Vice-President of the Student Union (BDE, 2024-2025) at IUT Salon and co-organizer of the GEII 2024 national symposium.

Education & Diplomas

- **2026 - Present:** Engineering Degree at ENSEEIHT (Electronics, Electrical Energy and Automation major - 3EA), Toulouse.
- **2023 - 2026:** Bachelor of Technology (BUT) in Electrical Engineering and Industrial Computing (GEII) (IUT Aix-Marseille, Salon-de-Provence | Ranked Top 3 of the graduating class).
- **2023:** High School Diploma (Baccalauréat) - Science Major (Maths & Physics) - Graduated with Honors (Lycée de Valabre).
- **Certifications:** Cisco CCNA (Networking), B2V Electrical Clearance, TOEIC (Score: 615), BIA (Aeronautics), BAFA and ASF (Youth worker/Animation), PSC1 (First Aid).